# Peer Review on Orion

November 1, 2025

## 1 Summary

This report discusses Fully Homomorphic Encryption (FHE) and how Orion, a recent framework, optimizes it. FHE is used for privacy-preserving machine learning, because it allows computation over encrypted data. When the computed ciphertext is decrypted, it corresponds to the same result as if the content had been decrypted first to plaintext and then calculated.
Strengths: well-explained and easy to follow introduction and background.
Weaknesses: at the moment still a bit short, lacking more in-depth information about the paper.

## 2 Clarity

The scheme CKKS is mentioned in the introduction, which is a bit confusing as it is not clear if this is a Abkürzung or just the name.

Overall, the report is clear and easy to understand. It provides a good overview of the paper studied and explains the most important topics in an understandable way.

## 3 Soundness

The background section 2.1 about Fully Homomorphic Encryption is really good explained, I understood everything the first time I read it, without having heard anything about this topic before. However, you could switch section 2.2 and 2.1, as the terms plaintext and ciphertext are only explained in section 2.2, but are already used in section 2.1. It might be more appropriate to remove section 2.4 on the Orion framework from the background and move it to a separate section, as it relates more to the main content.

In the lecture slides, it says you should have no section without text. You could add some sentences about what you are going to explain in the following chapters to avoid this.

As far as I understand, your own reproduction of the implementation does not belong in the report. You could move this section to the appendix and focus more on how the authors implemented Orion, for example. This would also mean that the table, which currently consists of many empty fields, would provide more information.

## 4 scholarship

The statements are well supported by sources. They are correctly and consistently cited and the links work. In the references, I could only find the link to Orion's GitHub. It would be good if the paper on Orion were also cited, as it is the most important source.

The lecture slides mention that the section on related work should not only list these, but also compare them. The section on related work provides a good overview of previous work on which Orion is based, as well as similar work. At the moment, however, it is still a little too much of a mere list. In addition, these works could be linked more closely with each other and with Express.

# 5 Minor comments

- Sometimes you have split up words in the middle of the line, like "mul- tiplication"
- The report is still a bit short with four sides, it should be 10-12 sides
- Little spelling error: "no problems <u>where</u> encountered"